

ADVANCED DATA MANAGEMENT 2025/2026
January 8, 2026
CLOSED BOOK

FIRST NAME:
SID:

LAST NAME:

Q1. (up to 10 lines) [**only for COMPUTER ENGINEERING STUDENTS**]
Briefly compare two of the replication protocols introduced in the course, highlighting the number of leaders, the execution of read and write operations, and the types of conflicts that may arise.

Q2. (up to 10 lines)
Explain why, in aggregate-oriented systems, uniqueness constraints can only be defined on a super-identifier.

Q3. (up to 10 lines)

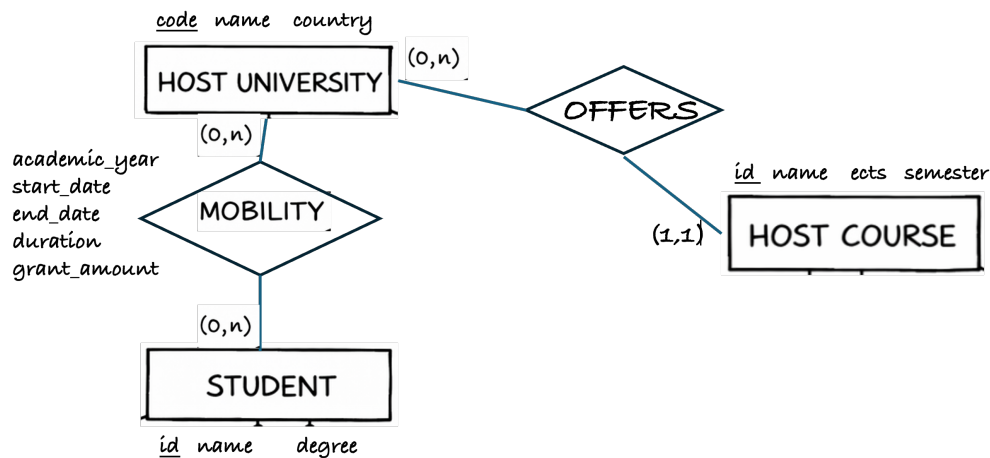
Briefly describe the three main approaches to query routing under consistent hashing (i.e., hashing on a ring), and discuss the number of nodes accessed in the worst case for each approach.

Q4. (up to 10 lines) [**only for COMPUTER SCIENCE STUDENTS**]

Discuss the risks of naïve post-filtering in a hybrid search system with vector similarity and metadata filtering (e.g., language = "Italian").

Exercise 1 [NoSQL aggregate-oriented data stores]

Consider the following entity relationship diagram, which describes the management of Erasmus student mobility at a university, including mobility assignments, learning agreements, and exam recognition.



The workload is the following:

- **Query 1:** Determine the courses offered by a university in France in the second semester.
- **Query 2:** Given a student identifier, retrieve the name and the degree of the student, together with the information about all her/his mobility experiences (academic year, start date, end date, code of the host university).
- **Query 3:** Determine the codes of all the host universities chosen for mobility by students of the “Computer Science” degree.

Based on the previous ER schema and workload, answer the following questions, **motivating all your choices**.

1. Use the aggregate-oriented design methodology (STEP 1-2-3) to design a set of logical aggregates, in meta-notation, for the given domain and the selected workload. Present the output of each step of the methodology.

Queries

Q1

Q2

Q3

Annotation: please annotate the input ER diagram

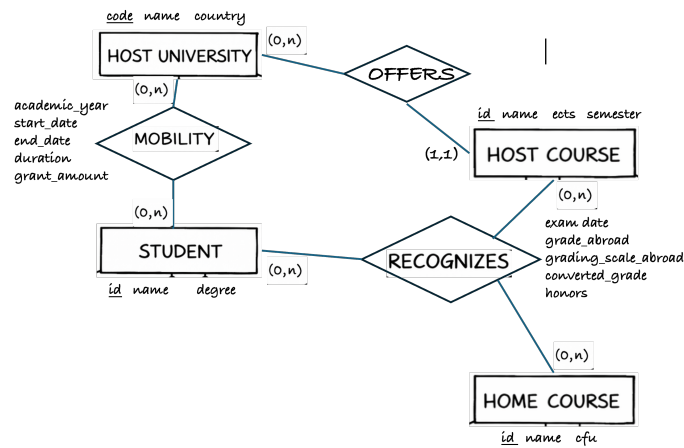
Aggregates

2. Revise and extend the logical aggregate schemas to consider partition keys and super-identifier information.
3. For each aggregate-oriented schema and each associated query, specify the query type (either batch or random-access) and whether it can be efficiently executed on the collection (for random-access queries only).
4. Determine which indexes (if any) should be created to improve the efficiency of workload queries.
5. Suppose to implement the designed aggregates in MongoDB.
 - a. Specify which indexes should be created and which of them should be unique.
 - b. Specify Query 3 in the MongoDB interaction language.

Exercise 2 [NoSQL graph-based data stores]

Consider the following entity relationship diagram, that describes the management of Erasmus student mobility at a university, including mobility assignments, learning agreements, and exam recognition:

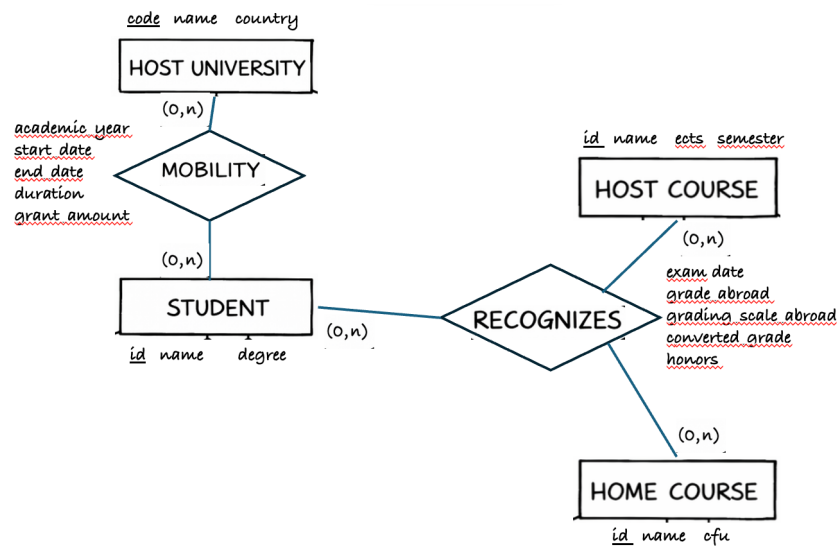
1. Present the most suitable logical schema in a graph-based data store relying on Labelled property graphs, like Neo4j (in the form of a graph pattern or of a sample instance), for the domain described by the Entity Relationship diagram, focusing on at least an instance for each relationship.



2. Formulate in Cypher the query to check that for all the recognized exams the host course is actually offered by the host university of the mobility and that the exam date is included in the mobility period. If the check succeeds, return the host course name, the exam date and the converted grade.

Exercise 3 [Data management for AI driven application] [only for COMPUTER SCIENCE STUDENTS]

With reference to the following domain describes the management of Erasmus student mobility at a university, including mobility assignments, learning agreements, and exam recognition:



1. Considering one small example instance of your choice (containing at least an instance of the relationships), discuss the differences in how the relationships are represented in a labelled property graph (e.g., Neo4j) and in RDF. Your answer should show the graphs, clearly explain and compare the modeling choices adopted in each case.

2. Assume now that the information of the Erasmus mobility domain is represented as a graph (e.g., a labelled property graph or an RDF graph). Discuss whether and how such a representation could be exploited in a Retrieval-Augmented Generation (RAG) setting involving Large Language Models (LLMs). In particular,
 - a) explain what advantages the use of an LLM provides compared to directly querying the graph using Cypher or SPARQL, and clarify the role of the textual component in this context.
 - b) Provide at least one concrete example of a question or task for which the graph could be used to retrieve structured information to support or constrain the LLM's output, and sketch the envisioned retrieval pipeline at a logical level.